

Modified Hurkx band-to-band-tunneling model for accurate and robust TCAD simulations

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ABSTRACT

Band-to-Band Tunneling (BTBT) is a non-local process and is responsible for some of the major leakage mechanisms in transistors such as Gate Induced Drain Leakage (GIDL). It is therefore important to include BTBT in Technology Computer Aided Design (TCAD) simulations to properly capture the leakage current. However, due to its non-local nature, BTBT simulation is computationally intensive. Local BTBT models such as Hurkx model can reduce the computation burden but often underestimate the leakage current at large bias. A new modified Hurkx BTBT model for homojunction devices is proposed and showed to match non-local BTBT model well in diode leakage and transistor GIDL simulations. Compared to non-local BTBT model, using the modified Hurkx model, the simulation speed is increased by > 6 times in a 3D FinFET I_DV_G simulation with GIDL.

1. Introduction

Reducing power consumption while increasing computing performance has been the main thrust to continue Moore's Law in the recent decade and is expected to continue [1]. It is therefore very important to model leakage mechanisms accurately in Technology Computer Aided Design, which has been used extensively by the industry to predict and design the next generation transistors [2]. One of the main leakage mechanisms in modern transistors is Band-To-Band-Tunneling [3], in which electron from valence band tunnels through the bandgap to conduction band, resulting in a generation of electron-hole pair under high electric field. One of the major leakage mechanisms BTBT is responsible for is Gate Induced Drain Leakage [4]. BTBT is also widely exploited to develop novel high I_{ON}/I_{OFF} ratio transistors such as Tunnel Field Effect Transistor (TFET) [5].

Due to its non-local nature (i.e. the electrons and holes are generated at different spatial locations), modelling BTBT is computationally intensive. This is because tunneling direction and electron and hole generation locations and rates depend on the band profile along the tunneling path and need to be computed and updated at every Newton iteration [6]. Therefore, local models in which electron/hole generations depend only on local electric field have been proposed to allow rapid computation of BTBT. Among them, Hurkx BTBT model is one of the most popular ones [6,7].

However, Hurkx model, like other local models, has limitations in predicting BTBT due to its local approximation. In particular, Hurkx

model underestimates the leakage current at large reverse bias. This can be alleviated by modifying the Hurkx equation to allow only generation but it will result in non-zero current at zero bias [6].

In this paper, we will discuss the limitation of Hurkx BTBT model and propose a local modified Hurkx BTBT model which matches non-local tunneling model well in 1D diode leakage, 2D planar MOSFET GIDL and 3D FinFET GIDL simulations. The simulation speed in 3D FinFET GIDL simulation is improved by 6 times using the modified Hurkx model over non-local BTBT model.

2. Issues in Hurkx model

Hurkx BTBT model, which is a local model, predicts the recombination rate (R) of electron-hole pair due to BTBT using the following equations [6,7]:

$$R = AD \left(\frac{F}{1 \text{ V/cm}} \right)^P e^{\frac{-B}{F}} \quad (1)$$

$$D = \frac{np - n_{i,eff}^2}{(n + n_{i,eff})(p + n_{i,eff})} (1 - |\alpha|) + \alpha \quad (2)$$

where F is the electric field strength (V/cm), n is the electron carrier density (cm⁻³), p is the hole carrier density (cm⁻³), n_{i,eff} is the intrinsic carrier concentration (cm⁻³). A, B, P are fitting parameters. α was introduced in TCAD Sentaurus [6] such that by specifying α = 0, the original Hurkx model will be obtained. By specifying α = -1 and

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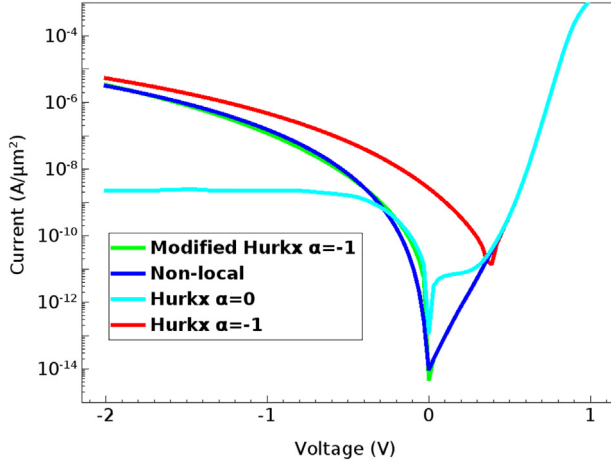


Fig. 1. Current-Voltage curve of silicon p-n diode with $N_A = 10^{20} \text{ cm}^{-3}$ and $N_D = 10^{19} \text{ cm}^{-3}$ using various BTBT models and setups. For the modified Hurkx model, $C = 0.5$ is used. Curves are essentially the same in forward region as BTBT has negligible effect.

$\alpha = 1$, it will result in generation-only ($D = -1$, suggested in [7] for large leakage current which will be discussed in more detail) and recombination-only ($D = 1$) process, respectively. In particular, when $\alpha = -1$, Eqs. (1) and (2) are simplified to,

$$R = -A \left(\frac{F}{1V/cm} \right)^p e^{-\frac{B}{F}} \quad (3)$$

and R is always negative, which means there is only electron-hole pair generation.

The issue of the original Hurkx model ($\alpha = 0$) is shown in Fig. 1. Fig. 1 shows the current-voltage (IV) relationship of abrupt p-n diode, with $N_A = 10^{20} \text{ cm}^{-3}$ and $N_D = 10^{19} \text{ cm}^{-3}$. While the original Hurkx model matches the non-local BTBT model well at low bias, at large reverse bias, it substantially underestimates the leakage current. This is because, in the original derivation, Eq. (2) is obtained by replacing neutral region electron and hole fermi levels by the local quasi-fermi levels of electrons and holes, respectively [7]. This also means the non-local nature of tunneling calculation becomes local. Such approximation breaks down when $np > n_{i,eff}^2$ due to significant electron-hole pair generations, during which local quasi-fermi levels are no longer correctly approximating the neutral region electron and hole fermi levels. Moreover, the local nature of this model also generates electrons and holes at the same spatial location (Fig. 2). When the generation rate is large, the error of the approximation is exacerbated as both n and p are large at the same spatial point and $(np - n_{i,eff}^2)$ can easily become positive in Eq. 2. In reality, electrons and holes are generated at different spatial locations (as shown by the non-local model in Fig. 2), so the recombination rate should be much lower in simulation and the leakage current should keep increases even at large reverse bias (Fig. 1).

To avoid the saturation of carrier generation in the original Hurkx model, one may use $\alpha = -1$ (i.e. $D = -1$ as suggested in [7]). By using $D = -1$, it is equivalent to treating the generation region as fully depleted of carriers ($n = p = 0$) regardless of the amount of electron hole pairs generated and this validates the approximation mentioned earlier in [7]. As shown in Figs. 1 and 2, the diode leakage current continues to increase as the reverse bias increases. However, it gives non-zero current at zero bias. This is because F is non-zero at zero bias in Eq. (3) due to built-in electric field. Or in other words, $D = -1$ is not valid under small reversed bias where n, p are not $\ll n_{i,eff}$.

3. Modified Hurkx model

To solve the problem in Hurkx model ($\alpha = 0$ or $\alpha = -1$), a

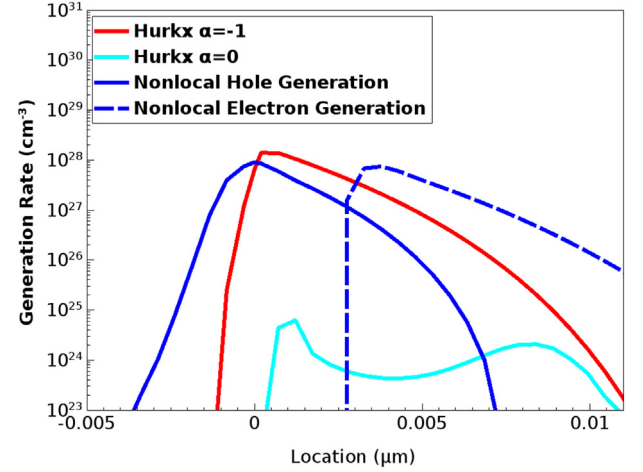


Fig. 2. Electron and hole generation rates along the diode in Fig. 1 at -2 V (reverse bias) with various models. The metallurgical junction is at 0 μm . p-region and n-region are at the negative and positive x-coordinates, respectively.

modified Hurkx Model is proposed by scaling Eq. (3) with a g -factor.

$$R = -gA \left(\frac{F}{1V/cm} \right)^p e^{-\frac{B}{F}} \quad (4)$$

$$g = \left(\frac{F - F_0}{F_0} \right)^{1.5} \text{ for } F \geq F_0 \text{ else } 0 \quad (5)$$

$$F_0 = \max \left(\tilde{F}_0, \sqrt{\frac{2qE_g N_{net}}{\epsilon}} \right) \quad (6)$$

where q is the electronic charge, E_g is the bandgap of the semiconductor, ϵ is the permittivity of the semiconductor, N_{net} is the net doping concentration, $\tilde{F}_0$ is the built-in electric field at zero bias. C is a fitting parameter.

The modified model is based on Eq. (3) because at large bias, $D = -1$ is a correct physical solution for a local model as discussed in Section 2. The form of g is chosen such that it gives zero generation at zero bias ($F - F_0 = 0$ in Eq. (5)) which solved the problem in the original Hurkx model with $\alpha = -1$. If $g = \left(\frac{F - F_0}{F} \right)$ would be used, the equation would reduce to the original Hurkx model with $\alpha = -1$. However, as showed in Fig. 1, the field dependence of the original Hurkx model with $\alpha = -1$ is not strong enough. Therefore, in the modified Hurkx model, $g = \left(\frac{F - F_0}{F_0} \right)^{1.5}$ is used instead, which gives a stronger F dependence as in the non-local model (Fig. 1). The second term in the max function of F_0 is used to estimate the built-in electric field of a pn junction. Setting $C = 0.5$ resembles the analytical solution of an abrupt diode peak electric field when $N_D = N_A = N_{NET}$ while setting $C = 1$ resembles the peak electric field of an abrupt diode with $N_D = N_{NET} \ll N_A$ (or vice versa). The max function is introduced for situations where the built-in electric field doesn't originate from a pn junction at equilibrium (e.g. originating from the gate work function and semiconductor Fermi level difference in a MOSFET, or due to fixed charges). In such cases, $F > \tilde{F}_0$ might not result in a band profile allowing BTBT as in a degenerately doped diode case. One can tune the C value to match non-local model results.

Fig. 3 compares the simulated IV curves of abrupt Silicon pn diode using non-local model and modified Hurkx model for various N_A and N_D combinations. $C = 0.5$ is used. Modified Hurkx model fits the non-local model very well.

The modified Hurkx model is calibrated to the non-local model and most of them are similar to the values given in [7].

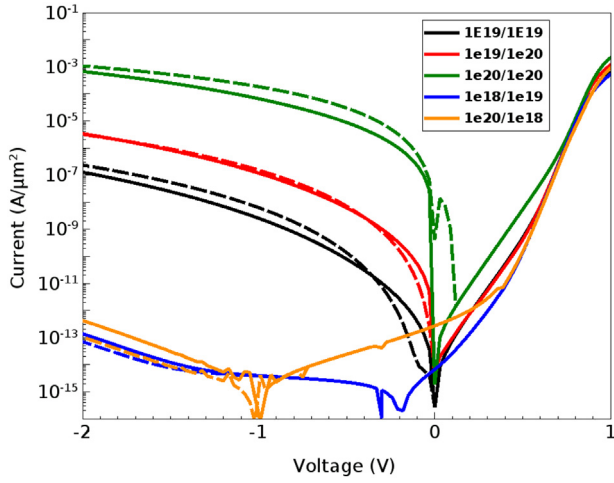


Fig. 3. Current-Voltage curves for silicon diodes with various N_D/N_A doping simulated with nonlocal model (dash lines) and modified Hurkx model (solid lines) with $C = 0.5$. Curves are essentially the same in forward region as BTBT has negligible effect.

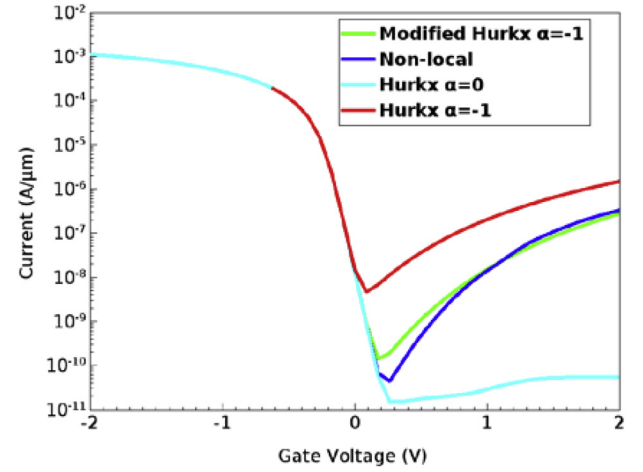


Fig. 6. $I_D V_G$ curves of 2D silicon pMOSFET with $V_D = 1.1$ V using various models for BTBT. Curves are essentially the same for $V_G < 0$ as BTBT has negligible effect.

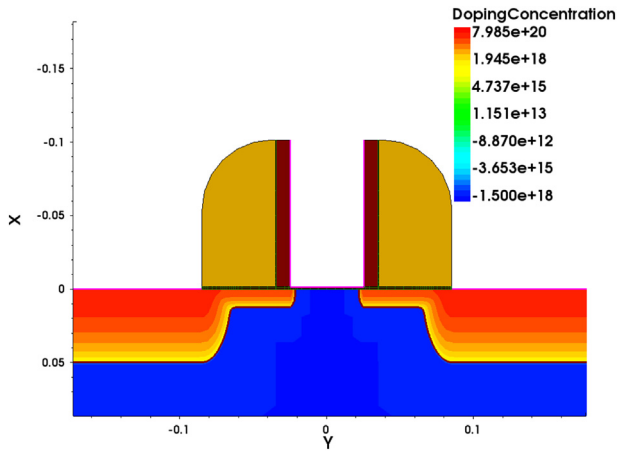


Fig. 4. 2D MOSFET structure used for GIDL simulation.

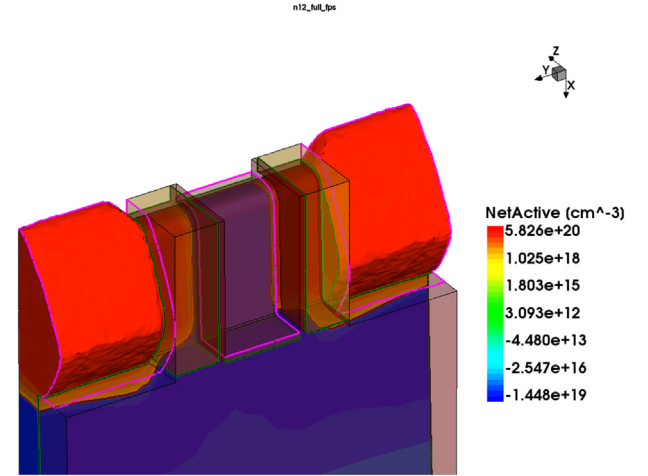


Fig. 7. 3D FinFET used in the simulation. Gate oxide and spacers are made transparent for clarity.

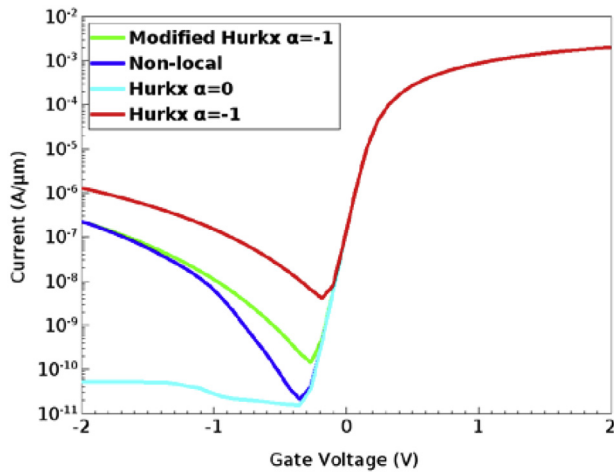


Fig. 5. $I_D V_G$ curves of 2D silicon nMOSFET with $V_D = 1.1$ V using various models for BTBT. Curves are essentially the same for $V_G > 0$ as BTBT has negligible effect.

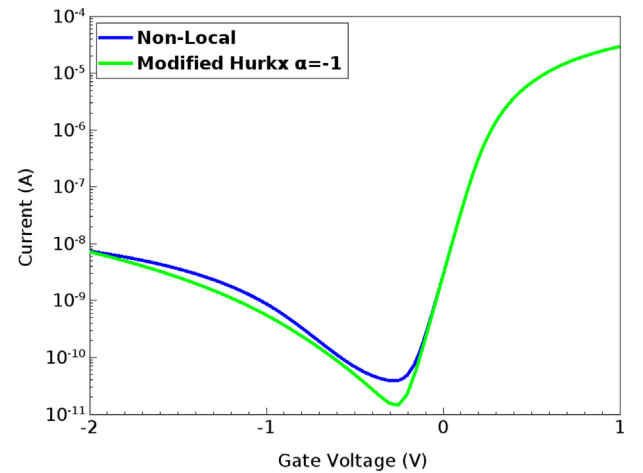


Fig. 8. $I_D V_G$ curves of 3D silicon nMOSFET with $V_D = 1$ V. Curves are essentially the same for $V_G > 0$ as BTBT has negligible effect.

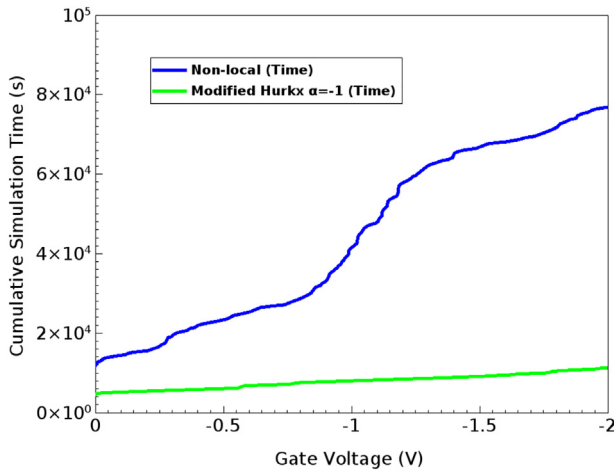


Fig. 9. Simulation time of 3D FinFET $I_D V_G$ simulation (when ramping V_G from 0 to -2 V) using modified Hurkx model and non-local model.

4. Modified Hurkx model for GIDL simulations

The modified Hurkx model is then used to simulate GIDL in 2D Silicon nMOSFET and pMOSFET. Fig. 4 shows the structure of the nMOSFET being simulated. The Equivalent Oxide Thickness (EOT) of the gate oxide is 1.2 nm with gate length (L_G) = 50 nm. Fig. 5 shows the $I_D V_G$ of the nMOSFET with drain voltage $V_D = 1.1$ V. The original Hurkx model with $\alpha = 0$ substantially underestimates the leakage current due to GIDL even at low negative gate bias while $\alpha = -1$ gives a leakage current that is more than two orders of magnitude larger than the non-local model at low negative gate bias. By using $C = 5$ in the modified Hurkx model, the leakage current due to GIDL can be modeled accurately as using the non-local model.

Similar results are obtained for 2D pMOSFET as shown in Fig. 6.

The modified Hurkx model is then applied in the $I_D V_G$ simulation of a realistic 3D Silicon FinFET structure shown in Fig. 7. The gate dielectric stack is composed of 2 nm HfO_2 and 1 nm Interfacial Oxide. L_g is 25 nm. Fin height is 40 nm and Fin width is 17 nm. Tensile stress is simulated in process [9] using Si:C stressor in the source-drain extension.

For realistic device simulation, the following models were included [6,8]: 1) Low-field mobility degradation due to high-k dielectric, 2) Coulomb scattering due to positive and negative charges at remote interfaces, 3) Density-gradient to account for quantum confinement effect, 4) Deformation potential model for the band structure, 5) mobility enhancement due to carrier redistribution in the bands, 6) Intel stress-induced electron mobility models, and mostly importantly, 7) BTBT model.

Fig. 8 shows that the modified Hurkx Model (with $C = 5$ as in the 2D case) matches the non-local model very well.

Fig. 9 shows the simulation statistics of the 3D FinFET $I_D V_G$ simulations with modified Hurkx and non-local models. Four cores were

used for each of the simulation. It can be seen that while modified Hurkx model gives similar GIDL current as non-local model in Fig. 8, it is 6 times faster than the non-local model. This is because in non-local BTBT model, tunneling direction and electron and hole generation locations and rates depend on the band profile along the tunneling path and need to be computed and updated at every Newton iteration.

5. Conclusion

Leakage simulation is becoming more important in TCAD due to the importance of reducing power consumption in transistors. BTBT is one of the major mechanisms of leakage in transistors. It is therefore important to have speedy and robust but accurate BTBT model incorporated in TCAD simulators.

In this article, a modified Hurkx model for homojunction devices is introduced. It is based on the original Hurkx model so the essential physics are preserved. It is a local model making it speedy and robust. It has a parameter to be calibrated to the non-local model (or experiment), which makes it simple to calibrate. It is showed that it matches the non-local BTBT model well in 1D diode leakage, 2D planar MOSFET GIDL and realistic 3D FinFET GIDL simulations. The simulation speed in a 3D FinFET GIDL simulation is improved by 6 times using the modified Hurkx model.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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